OPENSEARCH KERALA USER GROUP

Exploring OpenSearch

Core features and a hands-on demo

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Learn · Share · Build · Contribute

SPEAKER

Speaker introduction



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THIS SESSION

A map for your first hour with OpenSearch

01 **What it is**
The engine, and why a separate open-source project exists.

02 **The pieces**
The engine, Dashboards, Data Prepper, and plugins.

03 **Major features**
Search, analytics, observability, security, and vectors.

04 **The project**
The Foundation, and how a first contribution starts.

05 **A live demo**
One index, a search, an aggregation, then Dashboards.

THE PRODUCT

OpenSearch is a search and analytics engine

JSON in, answers out

You store documents, then search them or summarize them through a REST API.

Built on Apache Lucene

Lucene builds the inverted index that makes full-text search fast.

Distributed by design

Data is split across shards and copied to replicas, so a cluster can grow.

Apache License 2.0

You can use it, change it, and ship it. The project is community governed.

WHERE IT CAME FROM

The project joined the Linux Foundation in 2024

2021

OpenSearch is created from Elasticsearch 7.10.2, the last Apache 2.0 release, so search can stay open source. AWS hosts the project.

16 September 2024

The Linux Foundation launches the OpenSearch Software Foundation. AWS transfers OpenSearch into the OpenSearch Software Foundation.

OpenSearch Software Foundation

The OpenSearch Software Foundation is a Linux Foundation project. It is the vendor-neutral home of OpenSearch: infrastructure, events, legal support, and long-term sustainability.

Who decides

Inside the OpenSearch Software Foundation, the Governing Board looks after membership, budget, and outreach. The Technical Steering Committee makes the technical decisions.

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Five places you will meet OpenSearch

01

App search

Products, articles, or users behind a search box.

02

Log analytics

Central logs you can filter and count.

03

Observability

Logs, metrics, and traces in one system.

04

Security

Detect and investigate events in the same data.

05

Vector search

Similarity search for semantic search and RAG.

MAJOR PLUGINS

The product, and the major plugins

OpenSearch

The cluster. It stores documents and runs queries.

Dashboards

The UI for Discover, charts, and dashboards.

Data Prepper

A pipeline that receives data, changes it, and sends it in.

ML Commons

Run models and agent tools in the cluster.

Observability

Logs, traces, and application analytics.

Cross-Cluster Replication

Copy an index to another cluster.

k-NN

Nearest-neighbor search over vectors.

Neural Search

Semantic search using embeddings.

JVector

Disk-based vector search for very large indexes.

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SEARCH

Keyword, semantic, and hybrid search

Keyword search

Finds documents that contain the words you typed. BM25 ranks the closest matches.

EXAMPLE

"battery" matches "car battery replacement". It does not match "power backup".

Semantic search

Finds documents with a similar meaning, using vectors. Those exact words do not have to appear.

EXAMPLE

"power backup" can match "UPS and inverter for home use".

Hybrid search

Combines the keyword score and the semantic score, so exact words and meaning both count.

EXAMPLE

"rain coat" keeps the item named "rain coat" first, and can also include "waterproof jacket".

ONCE IT IS IN PRODUCTION

Observe it, protect it, and look after the data

Observability

Logs, metrics, and traces together. Trace analytics shows which service was slow.

Security

Authentication, roles, and permissions down to an index, a document, or a field.

Alerting

Run a query on a schedule and notify someone when the result matters.

Anomaly detection

Learn a normal pattern and flag values that do not fit it.

Index State Management

Roll over an index when it is large, and delete data when it is old.

Snapshots

Backups of the cluster. This is the recovery story.

Migration Assistant

Moves documents, mappings, templates, and aliases from Elasticsearch, OpenSearch, or Solr onto OpenSearch.

Backfill

Reads historical documents from a snapshot, so the source cluster stays quiet.

Capture and Replay

On supported versions, replays live writes so the cutover can have little or no downtime.

Then switch

Validate the target first. Example: Elasticsearch 7 to OpenSearch 3 in one step, then move traffic.

WHERE THE PROJECT IS GROWING

Vectors, models, and a UI on the same API

k-NN and hybrid search

Documents can store vectors. k-NN finds nearest neighbors. Hybrid search mixes keyword relevance with vector similarity.

Agentic search

Ask in plain language. An agent plans the query and runs it. Example: "Find red cars under $30,000."

ML Commons

Train and run models in the cluster, and connect agents to tools such as index search.

Dashboards

Discover is where you read documents. Visualizations turn aggregations into charts the team can watch.

HANDS-ON DEMO

One index. Three questions.

1 **Create**

An index of books. title is text, category is keyword, price is a number.

2 **Search**

Match a word in the title. Point at _score. That is relevance.

3 **Summarize**

A terms aggregation on category, in the same request as the search.

4 **Look**

Open the same search in Dashboards Discover, then one bar chart.



Run it locally

Demo files to launch OpenSearch

github.com/akolarkunnu/experiments



Free trial console

Create and manage a cluster

console2.instaclustr.com/login



Create a cluster

Free trial, step by step

https://www.instaclustr.com/support/documentation/opensearch/getting-started-with-opensearch/creating-an-opensearch-cluster/

DEMO BACKUP

The same three calls, if the cluster is shy

Create the index

```
PUT books
{ "mappings": { "properties": {
  "title":    { "type": "text" },
  "category": { "type": "keyword" },
  "price":    { "type": "float" } } } }
```

Search

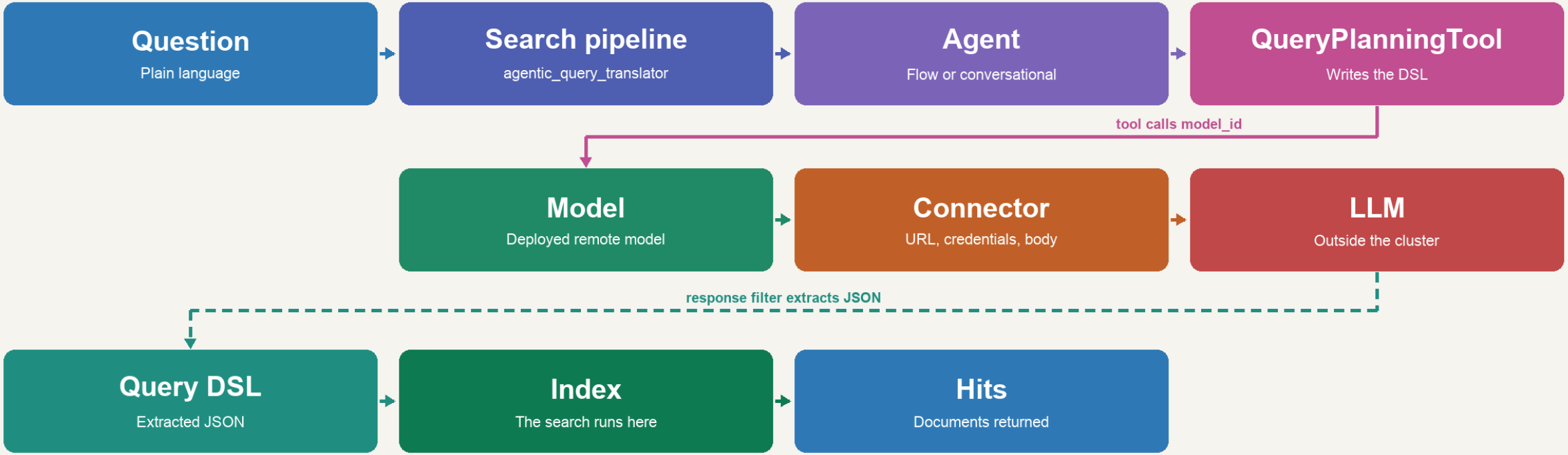
```
GET books/_search
{ "query": { "match": {
  "title": "search" } } }
```

Summarize

```
GET books/_search
{ "size": 0,
  "aggs": { "by_category": {
    "terms": { "field": "category" } } } }
```

HOW THE PARTS CONNECT

Agentic Search in OpenSearch



Demo follows this guide

Build your own chatbot

docs.opensearch.org/latest/tutorials/gen-ai/chatbots/build-chatbot/

Commands for this demo

The calls used in the walk-through

github.com/akolarkunnu/experiments

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TAKE THESE WITH YOU

Three sentences

1 Documents live in an index. A query finds them. An aggregation summarizes them.

2 One engine covers search, logs, observability, security, and vector search.

3 The project is Apache 2.0, hosted by the OpenSearch Software Foundation, and you can contribute.

CONTRIBUTION

A first change does not start in the core

1

Use it

Run a local cluster and follow one tutorial.

2

Talk

Ask in the forum, in Slack, or in this user group.

3

Pick an issue

Docs, a test, or a ticket marked good first issue.

4

Open a PR

Each commit needs a Signed-off-by line. That is the DCO.

How to start contributing to OpenSearch

A walk-through of a first contribution.

youtube.com/watch?v=Bvm1vJmZAcQ

COMMUNITY ROLES

Contributor, Member, Maintainer, Ambassador

Contributor

Open issues, send pull requests, and review other people's changes.

Anyone can start. Follow the Code of Conduct. No approval is required.

Member

Help run the queue: labels, assignees, and duplicate issues.

Sustained contributions, plus one maintainer who sponsors you.

Maintainer

Approve and merge changes, and guide the technical direction of one repository.

A sustained body of work in that repository, then a nomination.

Ambassador

Run meetups, write, and help newcomers get started.

Represent the community with respect. This user group is that work.

A vendor-neutral home, since September 2024

On 16 September 2024 the Linux Foundation launched the OpenSearch Software Foundation. AWS transferred the project into it. The Foundation supplies a neutral home, infrastructure, events, and legal support.

Governing Board

Membership, budget, and outreach.

Technical Steering Committee

Technical decisions stay here, not with the board.

Fifteen elected members. Each term is two years.

Where the global community meets

01

OpenSearchCon Europe

16–17 April 2026

Prague, Czechia

02

OpenSearchCon India

15–16 June 2026

Mumbai, India

03

OpenSearchCon NA

22–24 September 2026

San Jose, California

KEEP LEARNING

Two courses and one lab

InstaAcademy

COURSES

Advanced RAG for OpenSearch

Optimizing OpenSearch vector storage and search for faster AI applications

instaclustr.com/instacademy-courses/

Instaclustr Labs

LAB

Agentic RAG

github.com/instaclustr/labs

KEEP GOING

Product and community	opensearch.org
Guides and the REST API	docs.opensearch.org
Foundation and governance	opensearch.org/foundation
Code, issues, and first contributions	github.com/opensearch-project
Questions and design discussions	forum.opensearch.org
Free trial console	console2.instaclustr.com/login
Create a cluster	instaclustr.com/.../creating-an-opensearch-cluster/
Local OpenSearch demo	github.com/akolarkunnu/.../demo_materials
Build a chatbot	docs.opensearch.org/.../build-chatbot/
Agentic demo commands	github.com/akolarkunnu/.../demo_materials
Contributing video	youtube.com/watch?v=Bvm1vImZAcQ
InstaAcademy courses	instaclustr.com/instacademy-courses/
Instaclustr Labs	github.com/instaclustr/labs

Let’s build the OpenSearch community in Keralam together.

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Thank you

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